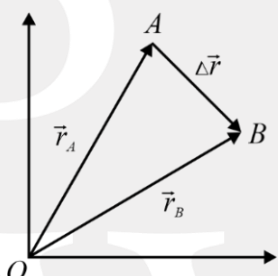
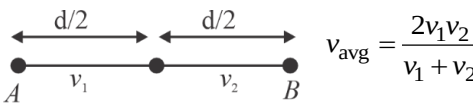
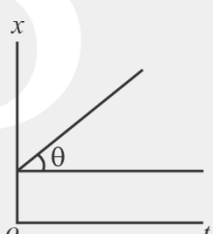
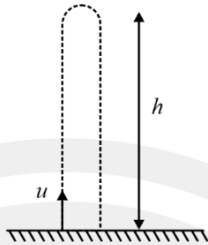
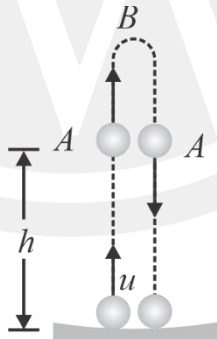


Motion in Straight Line

Sr. No.	Concept	Formulas	Other information
1.	Position vector ($\vec{r}$)	$\vec{r}_A = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$ $\vec{r}_B = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ 	Here, $\vec{r}_A(x_1, y_1)$ and $\vec{r}_B(x_2, y_2)$ are the position vectors.
2.	Displacement vector ($\Delta\vec{r}$)	$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$	Displacement vector $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$
3.	Average speed (v_{avg})	$\frac{\text{Total distance travelled}}{\text{Total time taken}} \quad \text{i.e.} \quad v_{\text{avg}} = \frac{\Delta s}{\Delta t}$	Δs = distance travelled Δt = time taken
4.	Instantaneous speed (v_{inst})	$v_{\text{inst.}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$	ds = short displacement dt = short time interval
5.	Average speed when particle moves with different uniform speeds in different time intervals	$v_{\text{avg}} = \frac{\text{Total distance covered}}{\text{Total time}}$ $= \frac{s_1 + s_2 + s_3 + \dots + s_n}{t_1 + t_2 + t_3 + \dots + t_n}$ $= \frac{v_1 t_1 + v_2 t_2 + v_3 t_3 + \dots}{t_1 + t_2 + t_3 + \dots}$ If $t_1 = t_2 = t_3 = \dots = t_n$ then $v_{\text{avg}} = \frac{v_1 + v_2 + \dots + v_n}{n}$	Different uniform speeds $v_1, v_2, v_3, \dots, v_n$ in different time intervals $t_1, t_2, \dots, t_n$ respectively

Sr. No.	Concept	Formulas	Other information								
6.	Average speed when particle describes equal distances with different speeds	$v_{\text{avg}} = \frac{d}{\frac{d}{nv_1} + \frac{d}{nv_2} + \dots + \frac{d}{nv_n}}$ For 2 equal distances. 	Particle moves a certain distance d in n equal parts with different speeds $v_1, v_2, v_3, \dots, v_n$								
7.	Average velocity	$\frac{\text{Net Displacement}}{\text{Time taken}}, \vec{v}_{\text{avg}} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{r}_f - \vec{r}_i}{t_f - t_i}$	$\vec{r}_f$ = final position $\vec{r}_i$ = initial position $t_f - t_i$ = time interval								
8.	Velocity in vector form	$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(x\hat{i} + y\hat{j} + z\hat{k}) = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$ $= v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$	Velocity can be positive, negative or zero.								
9.	Average acceleration	$\frac{\text{Total change in velocity}}{\text{Total time taken}} = \vec{a}_{\text{av}} = \frac{\Delta \vec{v}}{\Delta t}$	Δv = change in velocity Δt = time interval								
10.	Acceleration in vector form	$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(v_x\hat{i} + v_y\hat{j} + v_z\hat{k})$ $= \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$ $= a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$	dv = change in velocity dt = time interval								
11.	Instantaneous acceleration	$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} \quad \left[\text{As } \vec{v} = \frac{d\vec{r}}{dt} \right]$ $a_{\text{inst}} = \frac{dv}{dt} \Rightarrow a_{\text{inst}} = \frac{dv}{dx} \times \frac{dx}{dt}$ $a_{\text{inst}} = v \frac{dv}{dx} \quad \text{When } \vec{v} \text{ is function of position (x).}$	a_{inst} = instantaneous acceleration dx/dt = instantaneous velocity								
12.	Equations of motion	<table><tr><th>Scalar Form</th><th>Vector Form</th></tr><tr><td>$v = u + at$</td><td>$\vec{v} = \vec{u} + \vec{a}t$</td></tr><tr><td>$s = ut + \frac{1}{2}at^2$</td><td>$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$</td></tr><tr><td>$v^2 = u^2 + 2as$</td><td>$\vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + 2\vec{a} \cdot \vec{s}$</td></tr></table>	Scalar Form	Vector Form	$v = u + at$	$\vec{v} = \vec{u} + \vec{a}t$	$s = ut + \frac{1}{2}at^2$	$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$	$v^2 = u^2 + 2as$	$\vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + 2\vec{a} \cdot \vec{s}$	Here: v = final velocity s = displacement u = initial velocity t = time a = acceleration = constant These are valid only when acceleration is constant.
Scalar Form	Vector Form										
$v = u + at$	$\vec{v} = \vec{u} + \vec{a}t$										
$s = ut + \frac{1}{2}at^2$	$\vec{s} = \vec{u}t + \frac{1}{2}\vec{a}t^2$										
$v^2 = u^2 + 2as$	$\vec{v} \cdot \vec{v} = \vec{u} \cdot \vec{u} + 2\vec{a} \cdot \vec{s}$										

Sr. No.	Concept	Formulas	Other information
13.	Displacement in the n^{th} second (S_n)	$S_{n^{\text{th}}} = u + \frac{1}{2}a(2n - 1)$	u = initial velocity a = constant acceleration
14.	Galileo's law of odd numbers	When a particle starts from rest and moves with constant acceleration then the ratio of distance travelled by it in successive equal intervals of time is $1 : 3 : 5 : 7 : \dots (2n - 1)$	n = time interval in sec
15.	Stopping distance (s) & Stopping time (t)	$\Rightarrow s = \frac{u_0^2}{2a_0} \quad [\text{since } a_0 \text{ is constant}]$ $\Rightarrow t = \frac{u_0}{a_0} \quad [\text{since } a_0 \text{ is constant}]$ $\Rightarrow s \propto u_0^2 \quad \Rightarrow t \propto u_0$	u_0 = initial speed. a_0 = constant retardation
16.	Position-time graph ($x - t$)	The slope of the graph represents instantaneous velocity  $\therefore \tan \theta = \frac{\text{displacement}}{\text{time}} = \text{velocity}$	$\tan \theta$ = slope of the graph
17.	Velocity-time graph ($v - t$)	Slope of this graph represents acceleration. $\therefore \tan \theta = \frac{\text{velocity}}{\text{time}} = \text{acceleration}$ Area of $v - t$ graph = $\int v dt = \text{displacement} = \text{change in position}$	$\tan \theta$ = slope of the graph v = velocity dt = time interval
18.	Acceleration-time graph ($a - t$)	$\therefore$ Area of $a - t$ graph = $\int a dt = \int dv = v_2 - v_1 = \text{change in velocity}$	a = acceleration v_2 = final velocity v_1 = initial velocity
19.	Motion under gravity	If an object is falling freely ($u = 0$) (i) $v = u + gt$ (ii) $h = ut + \frac{1}{2}gt^2$ (iii) $v^2 = u^2 + 2gh$ If an object is thrown upward then g is replaced by $-g$ in the above three equations.	h = displacement of the body u = initial velocity t = time g = acceleration due to gravity v = final velocity

Sr. No.	Concept	Formulas	Other information
20.	A body is thrown vertically up with a velocity u	(i) Maximum height attained $h = \frac{u^2}{2g}$ (ii) Time of ascent = time of descent = $\frac{u}{g}$ (iii) Total time of flight = $\frac{2u}{g}$ 	h = maximum height u = initial velocity g = gravitational acceleration
21.	Effect of air resistance (Motion under gravity)	For ascending motion (upward motion) Time to go up, $t_1 = \sqrt{\frac{2h}{(g+a)}}$ Velocity of projection, $v_1 = \sqrt{2(g+a)h}$ For descending motion (downward motion) Time to go down, $t_2 = \sqrt{\frac{2h}{(g-a)}}$ Velocity on reaching ground, $v_2 = \sqrt{2(g-a)h}$	h = height attained by the body g = gravitational acceleration a = retardation due to air drag t_1 = time of ascent t_2 = time of descent
22.	If a body thrown upwards crosses a point in time t_1 & t_2 respectively then	 Height of point $h = \frac{1}{2} g t_1 t_2$ Maximum height $H = \frac{1}{8} g (t_1 + t_2)^2$ Time of flight = $t_1 + t_2 = \frac{2u}{g}$	t_1 = time of crossing a point while ascending t_2 = time of crossing the same point while descending

Sr. No.	Concept	Formulas	Other information
23.	Relative velocity	Relative velocity of A w.r.t. B $\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B$ Relative velocity of B w.r.t. A $\vec{v}_{B/A} = \vec{v}_B - \vec{v}_A$	$\vec{v}_A$ = velocity of A with respect to ground $\vec{v}_B$ = velocity of B with respect to ground

